

Tiny Basic Intermediate Language op-codes

yX	y0	y1	y2	y3	y4	y5	y6	y7	y8	y9	yA	yB	yC	yD	yE	yF
0x	SX x (7)								NO (6)	LB nn (8)	LN nnn n (11)	DS (8)	SP (7)	-	-	-
1x	SB (7- 9)	RB (7- 9)	FV (8)	SV (10)	GS (8)	RS (9)	GO (8+)	NE (8)	AD (14)	SU (14)	MP (14)	DV (31)	CP (10 - 11)	NX (20 +)	-	LS (40+)
2x	PN (42 - 44)	PQ (7+ 5n)	PT (7+ 2n)	NL (8)	PC a,” xx? ?”(5+5 n)	-	-	GL (16 ++)	-	-	IL (20 +)	MT (11)	XQ (12)	WS (15 0)	US (15- 20)	RT ()
3x	JS aa (9)								Jaa (7)							
4x																
5x																
6x																
7x																
8x	BC a, “xx??” (9+6n)															
9x																
Ax	BV a (10+)															
Bx																
Cx	BN a (15+5n)															
Dx																
Ex	BE a (7+)															
Fx																

Instruction groups:	Notes and examples:
Basic program control	Basic GOSUB is essentially GS+GO, and RETURN is RT, operating on separate Basic return stack. All in this group depend on the correct line structure in Basic memory.
IL program control	Only BR (branch relative) can branch “back”, while other branch instructions BC, BV, BN, BE only “forward”. CP (compare) is also a conditional skip and operates on evaluation stack.
Evaluation stack	16-bit literal value in LN (load number) is big-endian (most significant byte first). Evaluation stack has byte granularity, but most operations are 16-bit words.
Input / output	GL (get line) is a simple text input editor that operates on input buffer (128-character buffer in this implementation) and accepts input (also for the INPUT statement). All others in this group output characters.
Basic program memory manipulation	IL (insert line) maintains the Basic program structure in memory.
External	US implements USR function which has a number of predefined operations, most importantly PEEK and POKE.
-	Not implemented (execution will cause internal error and warm (re) start). May be used in future extended versions of Tiny Basic
Bolded instructions (BC – string match branch and PC – print literal) have variable length, 1+n bytes, with last byte marked with bit7 (MSB) set.	
<i>Italicized instructions</i> are multi-byte, 2 or 3. LN nnnn is the only 3-byte instruction.	
(value) in parenthesis is approximate execution time in micro-instructions. Each micro-instruction executes in 1 clock cycle, unless there is a wait (for example on I/O port or memory ready). Cost of instruction fetch (5 cycles) is included in all. Variations: • (from-to) - fixed range based on path taken • (..+xn) - x additional cycles for each byte traversed (e.g. PQ – print Basic string - takes 5 cycles to read from Basic memory and output for each character) • (..+) - any number of additional cycles, depending on data structure, but not indefinite. Examples are LS (List), IL (insert line), GO which heavily depend on how big the Basic program is	

- (..++) - any number of additional cycles, including infinite. The only example is GL (Get Line) which runs an infinite loop until ENTER or CTRL/C is pressed.